Correlations of the Thue-Morse Sequence

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Abstract. We revisit the pair correlations of the Thue–Morse sequence with focus on asymptotic results on various means. In the first part of the talk, we will show that all higher-order correlations of the Thue–Morse sequence with general real weights are effectively determined by a single value of the balanced 2-point correlation. As a consequence, we get that all odd-order correlations of the balanced Thue–Morse sequence vanish, and that, for any even n, the n-point correlations of the balanced Thue–Morse sequence have mean value zero, as do their absolute values, raised to an arbitrary positive power. We will then show how the correlations of the Thue–Morse sequence with general real weights can be derived from the balanced 2-point correlations. In the second/last part of the talk, we will focus on understanding the growth of the sum of the absolute values of the autocorrelations. This is joint work with Michael Baake, Jan Mazac, Ari Pincus-Kazmar and Adam Stout.

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27 October 2026

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Let $(t_k)_{k\geqslant 0}$ be the (one-sided) Thue–Morse sequence, or word, taking the values ± 1, defined by $t_0 = 1$ and, for $k \geqslant 0$, by

$$t_{2k} \;=\; t_k \quad \text{and} \quad t_{2k+1} \;=\; -t_k\,.$$

This sequence is the fixed point, starting from the seed a, of the morphism

$$\varrho \;=\; \varrho_{\mathrm{TM}} : \begin{cases} a \mapsto ab, \\ b \mapsto ba, \end{cases}$$

where we specialise the values by $a = -b = 1$.

We extend the Thue–Morse sequence to a bi-infinite sequence (or word) $w := (w_n)_{n\in\mathbb{Z}}$ by defining

$$w_n \;=\; \begin{cases} t_n\,, & \text{for } n \geqslant 0, \\ t_{-n-1}\,, & \text{for } n < 0. \end{cases}$$

The word w is the bi-infinite fixed point of the square of the Thue–Morse morphism, ϱ^2, starting from the seed $a|a$, where as before $a = -b = 1$.

Two point correlations via the Wiener Diagram

Considering just Thue–Morse sequence

$$\omega := \sum_{n\in\mathbb{Z}} w_n\,\delta_n \Longrightarrow \gamma_\omega := \omega \circledast \omega = \sum_{m\in\mathbb{Z}} \eta(m)\,\delta_m \Longrightarrow \widehat{\gamma}_\omega = \widehat{\omega \circledast \omega}$$

where ω is the (weighted) Dirac comb with weights w, $\circledast$ represents convolution and the values $\eta(m)$ are the autocorrelation coefficients

$$\eta(m) := \lim_{N\to\infty} \frac{1}{2N+1} \sum_{i=-N}^{N} w_i w_{i+m}.$$

Higher-order correlations via the dynamical system

Let S be the shift operator, $(Sw)_i = w_{i+1}$ and $\mathbb{X} = \mathbb{X}(w) := \overline{\{S^i w : i \in \mathbb{Z}\}} \subset \{\pm 1\}^{\mathbb{Z}}$ be the (discrete) hull.

- The space $\mathbb{X}$ together with the $\mathbb{Z}$-action of the shift forms a topological dynamical system $(\mathbb{X}, \mathbb{Z})$.
- It admits precisely one invariant probability measure, say μ, so $(\mathbb{X}, \mathbb{Z})$ is uniquely ergodic.
- Since $\mathbb{X}$ is also minimal (meaning that the $\mathbb{Z}$-orbit of every element is dense in $\mathbb{X}$), the system is even *strictly ergodic*.

Using this measure μ, and Birkhoff's ergodic theorem for uniquely ergodic shift spaces, one has

$$\int_{\mathbb{X}} x_{m_0} x_{m_1} \cdots x_{m_{n-1}} \, \mathrm{d}\mu(x) = \lim_{N \to \infty} \frac{1}{N} \sum_{k=0}^{N-1} y_{k+m_0} y_{k+m_1} \cdots y_{k+m_{n-1}},$$

where the choice of $y = (y_i)_{i \in \mathbb{Z}} \in \mathbb{X}$ is arbitrary.

Higher-order correlations via the dynamical system

Using this measure μ, and Birkhoff's ergodic theorem for uniquely ergodic shift spaces, one has

$$\int_{\mathbb{X}} x_{m_0} x_{m_1} \cdots x_{m_{n-1}} \, \mathrm{d}\mu(x) = \lim_{N\to\infty} \frac{1}{N} \sum_{k=0}^{N-1} y_{k+m_0} y_{k+m_1} \cdots y_{k+m_{n-1}},$$

where the choice of $y = (y_i)_{i\in\mathbb{Z}} \in \mathbb{X}$ is arbitrary.

Since the measure μ is shift invariant, without loss of generality, we can fix one of the m_i; we thus set $m_0 = 0$. We define the n-point correlations of the Thue–Morse sequence (and system) by

$$\begin{aligned} \eta(m_1, m_2, \ldots, m_{n-1}) &:= \int_{\mathbb{X}} x_0 \cdot (S^{m_1} x)_0 \cdots (S^{m_{n-1}} x)_0 \, \mathrm{d}\mu(x) \\ &= \lim_{N\to\infty} \frac{1}{N} \sum_{k=0}^{N-1} t_k t_{k+m_1} \cdots t_{k+m_{n-1}}. \end{aligned}$$

Patch frequency measure

We define the n-point correlations of the Thue–Morse sequence (and system) by

$$\begin{aligned}\eta(m_1, m_2, \ldots, m_{n-1}) &:= \int_{\mathbb{X}} x_0 \cdot (S^{m_1} x)_0 \cdots (S^{m_{n-1}} x)_0 \, \mathrm{d}\mu(x) \\ &= \lim_{N\to\infty} \frac{1}{N} \sum_{k=0}^{N-1} t_k \, t_{k+m_1} \cdots t_{k+m_{n-1}} .\end{aligned}$$

- μ is the *patch frequency measure* of the system, as defined by its values on the cylinder sets defined via all finite words.
- Their frequencies can be extracted from the frequency-normalised Perron–Frobenius eigenvectors of the induced substitution matrices for Thue–Morse words of length n.
- The frequency module—the Abelian group generated by all occurring frequencies—is $\mathcal{M}_\mu = \left\{ m/(3 \cdot 2^r) : r \in \mathbb{N}_0, m \in \mathbb{Z} \right\}$.

Two-point correlations

The standard two-point correlation coefficients $\eta(m)$ of the Thue–Morse sequence (and system), which are also known as the autocorrelation coefficients, are usually introduced as

$$\eta(m) \,=\, \lim_{N\to\infty} \frac{1}{N} \sum_{k=0}^{N-1} t_k \, t_{k+m},$$

which is consistent with our above definition. By symmetry, one has $\eta(-m) = \eta(m)$, which follows easily after dropping finitely many terms from the sum. We clearly get $\eta(0) = 1$, and, for $m \geqslant 0$, using the recursions of t_k, we have

$$\begin{aligned} \eta(2m) \,&=\, \eta(m) \qquad \text{and} \\ \eta(2m+1) \,&=\, -\frac{1}{2}\bigl(\eta(m) + \eta(m+1)\bigr). \end{aligned}$$

Two-point correlations

We have

$$\eta(2m) = \eta(m) \qquad \text{and}$$
$$\eta(2m+1) = -\frac{1}{2}\big(\eta(m) + \eta(m+1)\big).$$

These recurrences allow one to compute all of the values of η from $\eta(0)$,

$$\eta(1) = -\frac{\eta(0)}{3} = -\frac{1}{3},$$

We can write the pair of recurrence equations in the matrix form

$$\begin{pmatrix} \eta(2m) \\ \eta(2m+1) \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 2 & 0 \\ -1 & -1 \end{pmatrix}\begin{pmatrix} \eta(m) \\ \eta(m+1) \end{pmatrix},$$

which is valid for $m \geqslant 0$.

Two-point correlations and the nature of $\widehat{\gamma}$

Via the Wiener diagram, the coefficients $\eta(m)$ define a function $\eta\colon \mathbb{Z} \longrightarrow \mathbb{R}$, which

- is positive definite and defines a measure $\gamma = \eta \cdot \delta_{\mathbb{Z}}$
- γ is positive definite and pure point on $\mathbb{Z}$ so is Fourier transformable $\widehat{\gamma} = \nu_{\mathrm{TM}} * \delta_{\mathbb{Z}}$

It turns out ν_{TM} is a purely singular continuous probability measure on the torus $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, and has the Riesz product representation

$$\nu_{\mathrm{TM}} = \prod_{\ell=0}^{\infty} \bigl(1 - \cos(2^{\ell+1}\pi x)\bigr),$$

which converges weakly. Here, the right-hand side is the standard notation for a sequence of absolutely continuous measures, each given by its Radon–Nikodym density, with the limit being singular continuous.

The singular continuous nature of ν_{TM}

Theorem (Lebesgue decomposition)

Any regular Borel measure μ on $\mathbb{R}^d$ has a unique decomposition

$$\mu = \mu_{\mathrm{pp}} + \mu_{\mathrm{ac}} + \mu_{\mathrm{sc}}$$

where $\mu_{\mathrm{pp}} \perp \mu_{\mathrm{ac}} \perp \mu_{\mathrm{sc}}$ and also $|\mu| = |\mu_{\mathrm{pp}}| + |\mu_{\mathrm{ac}}| + |\mu_{\mathrm{sc}}|$.

The singular continuity of ν_{TM} rests upon two properties of η.

- By showing
$$\lim_{N\to\infty} \frac{1}{N} \sum_{m=0}^{N-1} \eta(m)^2 \,=\, 0,$$
one excludes pure point contributions, via Wiener's criterion.
- And, since $\eta(2^\ell) = -1/3$ for all $\ell \in \mathbb{N}$, one can apply the Riemann–Lebesgue lemma to show that no absolutely continuous component can exist.

Variants on $\lim_{N\to\infty} \frac{1}{N}\sum_{m=0}^{N-1} \eta(m)^2 = 0$

We have $\lim_{n\to\infty} \frac{1}{N}\sum_{m=0}^{N-1} \eta(m) = 0$.

Set $\Sigma(N) = \frac{1}{N}\sum_{m=0}^{N-1} \eta(m)$. As $N \to \infty$, one has $\Sigma(2N+1) = \frac{2N}{2N+1}\Sigma(2N) + O(N^{-1})$. Then

$$\begin{aligned}
\Sigma(2N) &= \frac{1}{2}\left(\frac{1}{N}\sum_{m=0}^{N-1}\eta(2m) + \frac{1}{N}\sum_{m=0}^{N-1}\eta(2m+1)\right) \\
&= \frac{1}{2}\left(\frac{1}{N}\sum_{m=0}^{N-1}\eta(m) - \frac{1}{2N}\sum_{m=0}^{N-1}\big(\eta(m)+\eta(m+1)\big)\right) \\
&= \frac{\Sigma(N)}{2} - \frac{\Sigma(N)}{4} - \frac{\Sigma(N)}{4} + \frac{\eta(0)-\eta(N)}{4N} \\
&= \frac{1-\eta(N)}{4N} = O(N^{-1})
\end{aligned}$$

since $|\eta(m)| \leqslant 1$ for all $m \in \mathbb{N}_0$.

Means for general powers and $|\eta(m)|$

Proposition

For $k \in \mathbb{N}_0$, one has $\lim_{N\to\infty} \frac{1}{N} \sum_{m=0}^{N-1} \eta(m)^k = \delta_{k,0}$. Furthermore, one has $\lim_{N\to\infty} \frac{1}{N} \sum_{m=0}^{N-1} |\eta(m)|^k = 0$ for all $k \geqslant 2$. □

So, what about the mean of $|\eta(m)|$?

Via the Cauchy–Schwarz inequality in conjunction with $|\eta(m)| \leqslant 1$,

$$\left(\frac{1}{N} \sum_{m=0}^{N-1} |\eta(m)| \right)^2 \leqslant \frac{1}{N} \sum_{m=0}^{N-1} \eta(m)^2 \leqslant \frac{1}{N} \sum_{m=0}^{N-1} |\eta(m)|,$$

so

$$\lim_{N\to\infty} \frac{1}{N} \sum_{m=0}^{N-1} |\eta(m)| = 0.$$

Not only is the mean value of $|\eta(m)|$ equal to zero, we can do better.

Sub-linear growth of $\sum_{m=0}^{N-1} |\eta(m)|$

Theorem
For any $\alpha > \frac{\log(3)}{\log(4)} \approx 0.792481505$, *we have*

$$\lim_{N\to\infty} \frac{1}{N^\alpha} \sum_{m=0}^{N-1} |\eta(m)| \,=\, 0.$$

Note that, when $N \in [2^{2\ell}, 2^{2\ell+2}]$, one has the estimate

$$\frac{1}{N^\alpha} \sum_{m=0}^{N-1} |\eta(m)| \leqslant \frac{1}{(2^{2\ell})^\alpha} \sum_{m=0}^{2^{2\ell+2}-1} |\eta(m)| = \frac{4^\alpha}{(2^{2\ell+2})^\alpha} \sum_{m=0}^{2^{2\ell+2}-1} |\eta(m)|,$$

which implies that it suffices to show $2^{-(2\ell+2)\alpha} \sum_{m=0}^{2^{2\ell+2}-1} |\eta(m)| = o(1)$ as $\ell \to \infty$.

Proof of $\sum_{m=0}^{N-1} |\eta(m)| = o(N^{\log_4(3)})$ continued.

To this end, note that the recurrences (9) imply the estimates

$$
\begin{aligned}
|\eta(4m)| &= |\eta(m)|, \\
|\eta(4m+1)| &= \frac{1}{2}\big|\eta(2m)+\eta(2m+1)\big| = \frac{1}{2}\big|\eta(m)-\frac{1}{2}\big(\eta(m)+\eta(m+1)\big)\big| \\
&\leqslant \frac{1}{4}|\eta(m)|+\frac{1}{4}|\eta(m+1)|, \\
|\eta(4m+2)| &= |\eta(2m+1)| \leqslant \frac{1}{2}|\eta(m)|+\frac{1}{2}|\eta(m+1)|, \\
|\eta(4m+3)| &= \frac{1}{2}\big|\eta(2m+1)+\eta(2m+2)\big| \\
&= \frac{1}{2}\big|\eta(m+1)-\frac{1}{2}\big(\eta(m)+\eta(m+1)\big)\big| \\
&\leqslant \frac{1}{4}|\eta(m)|+\frac{1}{4}|\eta(m+1)|.
\end{aligned}
$$

Proof of $\sum_{m=0}^{N-1} |\eta(m)| = o(N^{\log_4(3)})$ continued.

Let $\alpha > \frac{\log(3)}{\log(4)}$ and set $\Sigma'(N) = \frac{1}{N^\alpha} \sum_{m=0}^{N-1} |\eta(m)|$. Arguing as above, we have

$$\Sigma'(4N) = \frac{1}{(4N)^\alpha} \sum_{r=0}^{3} \sum_{m=0}^{N-1} |\eta(4m+r)|$$

$$\leqslant \frac{1}{4^\alpha} \Bigg(\frac{1}{N^\alpha} \sum_{m=0}^{N-1} |\eta(m)| + \frac{1}{4N^\alpha} \sum_{m=0}^{N-1} \big(|\eta(m)| + |\eta(m+1)|\big)$$

$$+ \frac{1}{2N^\alpha} \sum_{m=0}^{N-1} \big(|\eta(m)| + |\eta(m+1)|\big) + \frac{1}{4N^\alpha} \sum_{m=0}^{N-1} \big(|\eta(m)| + |\eta(m+1)|\big) \Bigg)$$

$$= \frac{3}{4^\alpha} \Sigma'(N) + \frac{|\eta(N)| - |\eta(0)|}{4N^\alpha} \leqslant \frac{3}{4^\alpha} \Sigma'(N),$$

since $|\eta(N)| - |\eta(0)| = |\eta(N)| - 1 \leqslant 0$.

Proof of $\sum_{m=0}^{N-1} |\eta(m)| = o(N^{\log_4(3)})$, finished.

Finally, letting $N = 2^{2\ell+2}$, for $\ell \geqslant 0$, we get

$$\frac{1}{2^{(2\ell+2)\alpha}} \sum_{m=0}^{2^{2\ell+2}-1} |\eta(m)| = \Sigma'(2^{2\ell+2}) \leqslant \frac{3}{4^\alpha}\, \Sigma'(2^{2\ell}) \leqslant \left(\frac{3}{4^\alpha}\right)^{\ell+1} = o(1).$$

The $\log(3)/\log(4)$ is reminiscent of a result of Newman.

Theorem (Newman and Slattery, 1975)
There is a $C > 0$ such that

$$\sup_{\theta\in[0,1]} \left| \sum_{k\leqslant x} \mathrm{e}^{2\pi \mathrm{i} k\theta}\, t_k \right| \leqslant C x^{\log(3)/\log(4)}.$$

(The exponent $\log(3)/\log(4)$ is optimal.)

The optimal α?

So, how good can we do? What is the optimal α?

Using the Newman and Slattery result and a heuristic argument we expect that there is a $C > 0$ such that

$$\sup_{\theta\in[0,1]} \left| \sum_{m\leqslant x} \mathrm{e}^{2\pi \mathrm{i} m\theta} \eta(m) \right| \leqslant C x^{2\log(3)/\log(4)-1},$$

where $2\log(3)/\log(4) - 1 = 0.58496250072\cdots$.

Theorem (C., Mazáč, Pincus-Kazmar and Stout, 2026)
There exists positive constants c_1 and c_2, such that, for any x,

$$c_1 x^{0.6274882485} \leqslant \sum_{m\leqslant x} |\eta(m)| \leqslant c_2 x^{0.6464616661}.$$

Return to higher-order correlations

Recall, we define the n-point correlations of the Thue–Morse sequence (system) by

$$\begin{aligned}\eta(m_1, m_2, \ldots, m_{n-1}) &:= \int_{\mathbb{X}} x_0 \cdot (S^{m_1}x)_0 \cdots (S^{m_{n-1}}x)_0 \,\mathrm{d}\mu(x) \\ &= \lim_{N\to\infty} \frac{1}{N} \sum_{k=0}^{N-1} t_k t_{k+m_1} \cdots t_{k+m_{n-1}}.\end{aligned}$$

Proposition

For each $i \in \{1, \ldots, n-1\}$, let $r_i \in \{0, 1\}$ and set $r = r_1 + \cdots + r_{n-1}$. Then, for any integers $m_1, \ldots, m_{n-1} \geqslant 0$, we have

$$\eta(2m_1+r_1, \ldots, 2m_{n-1} + r_{n-1}) = \frac{(-1)^r}{2}\big(\eta(m_1, \ldots, m_{n-1}) + (-1)^n \eta(m_1 + r_1, \ldots, m_{n-1} + r_{n-1})\big).$$

Proof of Proposition (for those that want it)

Proof.

By Birkhoff's ergodic theorem in conjunction with the absolute convergence and hence rearrangement invariance of all involved sums, we have

$$\eta(2m_1 + r_1, \ldots, 2m_{n-1} + r_{n-1}) = \lim_{N\to\infty} \frac{1}{2N} \sum_{k=0}^{2N-1} t_k\, t_{k+2m_1+r_1} \cdots t_{k+2m_{n-1}+r_{n-1}}$$

$$= \lim_{N\to\infty} \frac{1}{2N} \sum_{j=0}^{N-1} \big(t_{2j}\, t_{2j+2m_1+r_1} \cdots t_{2j+2m_{n-1}+r_{n-1}} + t_{2j+1}\, t_{2j+2m_1+r_1+1} \cdots t_{2j+2m_{n-1}+r_{n-1}+1} \big)$$

$$= \lim_{N\to\infty} \frac{(-1)^r}{2N} \sum_{j=0}^{N-1} \big(t_j\, t_{j+m_1} \cdots t_{j+m_{n-1}} + (-1)^n\, t_j\, t_{j+m_1+r_1} \cdots t_{j+m_{n-1}+r_{n-1}} \big)$$

$$= \frac{(-1)^r}{2} \big(\eta(m_1, \ldots, m_{n-1}) + (-1)^n \eta(m_1 + r_1, \ldots, m_{n-1} + r_{n-1}) \big),$$

which completes the argument. □

Artistic digression: 4-point correlations

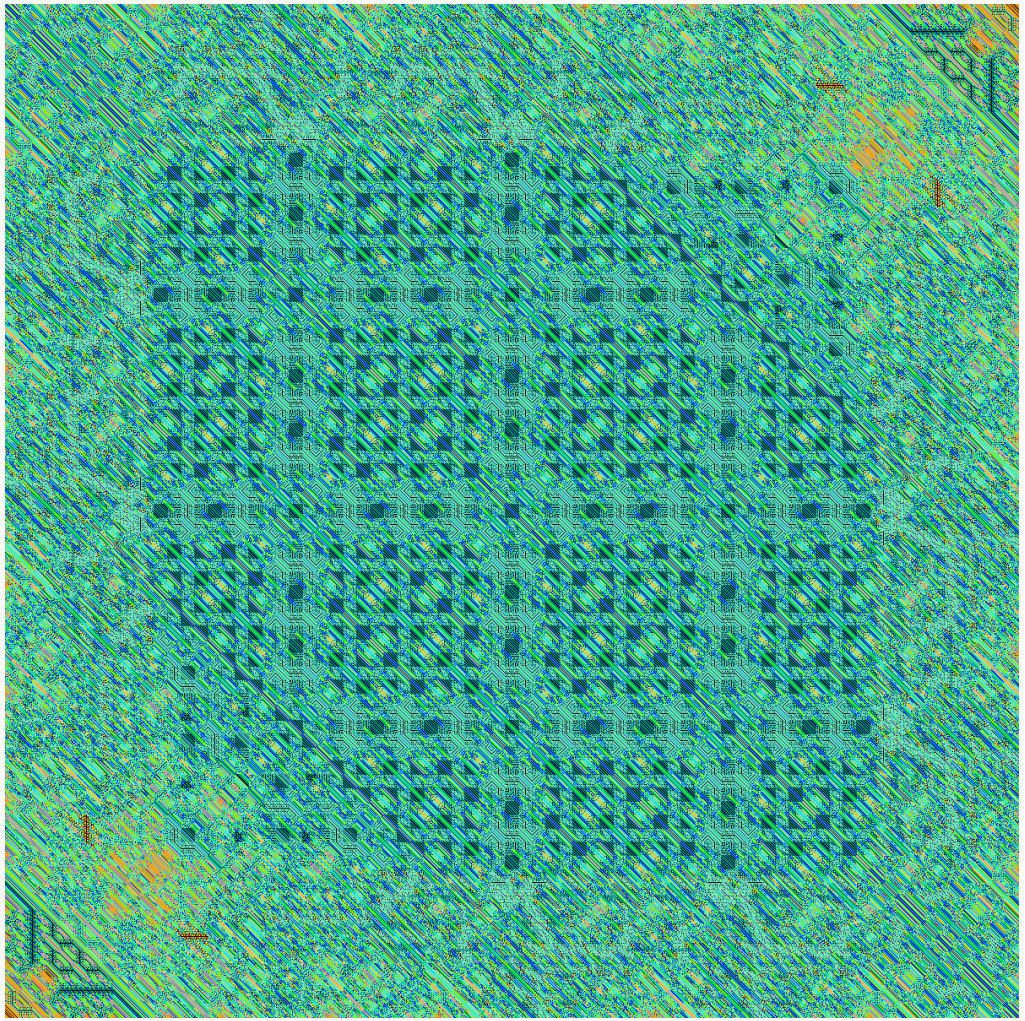


Figure 1: See "Abstract art generated by the Thue–Morse correlation functions" by Darren Ong.

Higher-order correlations

To compute all of the n-point correlations, we only need to know the values at the 2^{n-1} corners of the $(n-1)$-dimensional unit hypercube.

For a point $(r_1, \ldots, r_{n-1}) \in \{0,1\}^n$, with $r = r_1 + \ldots + r_{n-1}$,

$$\eta(r_1, r_2, \ldots, r_{n-1}) = \lim_{N\to\infty} \frac{1}{N} \sum_{k=0}^{N-1} t_k \, t_{k+r_1} \cdots t_{k+r_{n-1}}$$

$$= \begin{cases} 0, & \text{if } n \text{ is odd,} \\ 1, & \text{if } n \text{ is even and } r \text{ is even,} \\ -\frac{1}{3}, & \text{if } n \text{ is even and } r \text{ is odd.} \end{cases}$$

For example, if n is odd, then

$$\eta(r_1, r_2, \ldots, r_{n-1}) = \lim_{N\to\infty} \frac{1}{N} \sum_{k=0}^{N-1} t_k^{n-r} t_{k+1}^{r} = \lim_{N\to\infty} \frac{1}{N} \sum_{k=0 \text{ or } 1}^{N-1 \text{ or } N} t_k = 0.$$

Corollary

All odd-order correlations of the balanced Thue–Morse system vanish. □

Corollary

The solution space of the linear recursion equations for n-point correlations is one-dimensional. Consequently, for any $n \in \mathbb{N}$, the n-point correlations of the balanced Thue–Morse system are uniquely specified by a single number, namely by $\eta(0, \ldots, 0)$.

Proof.

All coefficients are fully determined once $\eta(r_1, \ldots, r_{n-1})$ is known for all $r_i \in \{0, 1\}$. Choosing all $m_i = 0$ in the recursion relations, one gets

$$\eta(r_1, \ldots, r_{n-1}) = \frac{(-1)^r}{2} \left(\eta(0, \ldots, 0) + (-1)^n \eta(r_1, \ldots, r_{n-1}) \right)$$

with $r = r_1 + \ldots + r_{n-1}$ as before. This simply gives

$$\eta(r_1, \ldots, r_{n-1}) = \frac{(-1)^r}{2 + (-1)^{n+r-1}} \, \eta(0, \ldots, 0) \tag{1}$$

for all $(r_1, \ldots, r_{n-1}) \in \{0, 1\}^{n-1}$, which establishes our claim. □

Means of higher-order correlations

Theorem

Let $n \geqslant 2$ be fixed. Then, for any $\alpha > \frac{\log(3)}{\log(4)}$ and $\beta > 0$, one has

$$\lim_{N\to\infty} \frac{1}{N^{\alpha(n-1)}} \sum_{0\leqslant m_1,\ldots,m_{n-1}\leqslant N-1} |\eta(m_1,\ldots,m_{n-1})| = 0 \quad \textit{and}$$

$$\lim_{N\to\infty} \frac{1}{N^{n-1}} \sum_{0\leqslant m_1,\ldots,m_{n-1}\leqslant N-1} |\eta(m_1,\ldots,m_{n-1})|^{\beta} = 0.$$

Means of higher-order correlations via regular sequence

One can study the correlations $\eta(m_1, \ldots, m_{n-1})$ via a related regular sequence. We have a general relationship,

$$\boldsymbol{\eta}(2m_1 + r_1, \ldots, 2m_{n-1} + r_{n-1}) = \boldsymbol{B}_i \boldsymbol{\eta}(m_1, \ldots, m_{n-1}), \tag{2}$$

where for each $i \in \{0, 1, \ldots, 2^{n-1} - 1\}$ with binary expansion $(i)_2 = r_1 r_2 \cdots r_{n-1}$, we have set

$$\boldsymbol{B}_i = \boldsymbol{B}_{(r_1, r_2, \ldots, r_{n-1})}.$$

Now, we define the sequence $\eta_n := (\eta_n(m))_{m \geqslant 0}$ by

$$\eta_n(m) = \boldsymbol{e}_1^T \boldsymbol{B}_{i_0} \boldsymbol{B}_{i_1} \cdots \boldsymbol{B}_{i_s} \boldsymbol{e}_1,$$

where $(m)_n = i_s \cdots i_1 i_0$ is the base-n expansion of m, and $\boldsymbol{e}_1$ is the standard column basis vector of length 2^n with the 1 in the first position.

Means of higher-order correlations via regular sequence

Theorem

Let $n \geqslant 2$. Then, η_n has mean value zero, that is,

$$\lim_{m\to\infty} \frac{1}{m} \sum_{j=0}^{m-1} \eta_n(j) = 0.$$

Here, instead of taking the mean value over the n-dimensional integer cubes of the first orthant as above, we traverse the integer points in that orthant according to the order imposed by the regular sequence.

References

M. Baake and M. Coons, *Correlations of the Thue–Morse sequence*, Special Issue on Aperiodic Order in honor of Uwe Grimm, **Indagationes Mathematicae 35** (2024), no. 5, 914–930.

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Tak, Thank you.

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27 October 2026

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